Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/26/2005
PI Name:	Revankar, Shripad T Ph.D.		
Project Title:	Study of Co-Current and Counter-Current Gas-Liquid Two-Phase Flow Through Packed Bed in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	shripad@ecn.purdue.edu	**Fax:**	FY 765-494-9570
PI Organization Type:	UNIVERSITY	**Phone:**	765-496-1782
Organization Name:	Purdue University		
PI Address 1:	400 Central Drive,		
PI Address 2:	Nuclear Enginering		
PI Web Page:	http://helios.ecn.purdue.edu/		
City:	West Lafayette	**State:**	IN
Zip Code:	47907	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	05/01/2002	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
Contact Email:	brian.j.motil@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2715		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The main goal of the project was to develop experiments to obtain data and develop models on the gas-liquid two-phase flow through a packed bed in microgravity and characterize the flow regime transition, pressure drop, void and interfacial area distribution, and liquid hold up. An experimental and analytical modeling program for the study of co-current air-water two-phase down flow in a packed column was developed. A detailed survey of the literature was performed on two-phase flow in packed columns. The literature shows only one set of study from Glenn Research Center NASA is available on two-phase flow in packed bed under microgavity conditions. A packed bed test facility was designed and constructed. The bed is a cylindrical pipe of diameter 69 mm and height of 1.52m. the bed particles were glass spheres of diameter 6 mm. The bed porosity was 0.377. Single phase flow characterization of the test loop was performed by flowing air and water in the bed and measuring bed pressure drop. From these measurements single phase constants in Ergun pressure drop equations were calculated. A large amount of effort was put in the development of instrumentation for the local and global measurement in the packed bed. A two point conductivity probe for the measurement of void fraction in the bed at pore level, a film thickness probe for the measurement of film thickness and drying characteristics of the packing, and a bed wise impedance meter for bed wise void measurement were developed. These probes were calibrated and were installed in the packed bed. Two Phase co-current down flow of air-water test were conducted. The flow regime in the bed was observed and recorded with high speed video camera. Flow regime map was developed. Five flow regimes were identified, trickling, wavy, transitional, pulsing and bubble flow. Pressure drop in the bed were also obtained for various flow regimes in the bed. An analytical model was developed for predicting flow regime transition from trickle to pulse mode using two-phase film model. Experimental flow regime identification method was developed using the signals from the film thickness probes and the bed void fraction data. Packed bed reactors have potential space-based applications in bioregenerative food production, waste treatment, watering and nutrient transport in plant root systems, waste recovery, and extraction from planetary materials. The present research addresses the most important topic of HEDS-specific microgravity fluid physics research identified by NASA ‘s one of the strategic enterprises, OBPR Enterprise. The first two years were devoted to ground based flight definition experimental and modeling program. In the current third year, detailed measurements are being carried out on various flow dynamics including pulse frequency. During the next year microgravity flight tests will be carried out using the ground-based parabolic flight research aircraft. The specific objectives of the proposed research are to: 1. Develop experiments for the study of the gas liquid two-phase flow through the packed bed with three different flow combinations: co-current down flow, co-current upflow and counter current flow. 2. Develop pore scale and bed scale two-phase instrumentation for measurement of flow regime transition, void distribution and gas-liquid interfacial area concentration in the packed bed. 3. Obtain database on flow regime transition, pressure drop, void distribution, interfacial area concentration and liquid hold up as a function of bed characteristics such as bed particle size, porosity, and liquid properties such as viscosity and surface tension. 4. Develop mathematical model for flow regime transition, void fraction distribution and interfacial area concentration for co-current gas-liquid flow through the porous bed in gravity and micro gravity conditions. 5. Develop mathematical model for the flooding phenomena in counter-current gas-liquid flow through the porous bed in gravity and micro gravity conditions. 6. Train graduate and undergraduate students in instrumentation and measurement methods with particular emphasis on two-phase flow through packed beds. Packed bed reactors are probably the oldest and most widely used catalytic chemical reactors. Packed beds are used for both single phase and multi-phase catalytic processes. Gas-liquid two-phase flow through a fixed bed of particles occurs in many unit operations of interest to the designers of space-based as well as terrestrial equipment. Examples include separation columns, gas-liquid reactors, humidification, drying and gas absorption operations and extraction and leaching of minerals. Trickle-bed reactors, in which a gas and a liquid flow cocurrently downward over a randomly packed stationary bed, are widely used in variety of gas-liquid -solid catalytic reactions including petroleum and petrochemical industries. Potential space-based applications include: fuel cells, bioregenerative food production, waste treatment, watering and nutrient transport in plant root systems, waste recovery, and extraction from planetary materials. In general, gravitational, capillary, and viscous forces drive flow in porous media. Gravity causes phase separation and migration in the direction of the gravitational field. In microgravity, capillary and viscous forces play fundamental role in controlling phase distribution and hence, multiphase flow and transport in packed beds. Flow regime, pressure drop, flow stability and dryout, therefore, are also expected to be markedly different in microgravity. Though recently various studies have been conducted for two-phase flow through packed bed at earth gravity condition to understand the mechanisms governing the flow regime transition, pressure drop, void and interfacial area distribution, there is no published work available on this topic at microgravity condition. Also the pore level study of the two-phase flow through packed beds has not been investigated in detail even at earth gravity condition. The bed characterization is generally done through the macro scale parameters such as total bed pressure drop, liquid hold up and total void. It should be noted that the macro scale parameters are rooted in the micro scale, where two-phases compete for the void space. In view of this an experimental and analytical study is carried out where systematic experiments are conducted for the gas-liquid two-phase flow in a packed bed in earth gravity and microgravity conditions. Data on key bed flow parameters are obtained both at micro scale and macro scale. Modeling efforts take account of the two-phase hydrodynamics in micro scale (pore) level. It is expected that the results of this modeling effort will eventually lead to better understanding of the bed characteristics at the microgravity conditions and hence support safe and efficient packed bed designs for space applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A packed bed test facility was designed and constructed. Single phase flow characterization of the test loop was performed by flowing air and water in the bed and measuring bed pressure drop. From these measurements single phase constants in Ergun pressure drop equations were calculated. Two Phase concurrent down flow of air-water test were conducted. The flow regime in the bed was observed and recorded with high speed video camera. Flow regime map was developed. Five flow regimes were identified, trickling, wavy, transitional, pulsing and bubble flow. Pressure drop measurements in the bed were also obtained for various flow regimes in the bed. These are traditional data for a packed bed at global level. They can provide credible data with high responsive speed. FFT and PDF were calculated from original data and different flow regimes show different profile shapes. Frequencies extracted by FFT can characterize the two-phase flow and show some phenomena which can not be displayed by original signal. Pulse frequency was calculated by the FFT method. Pulse velocity was calculated from the high speed camera images. The hydrodynamic properties of pulses in packed bed were concluded. New instruments with unique applications to packed bed were developed. A two point conductivity probe for the measurement of void fraction in the bed at pore level, a film thickness probe for the measurement of film thickness at pore level, and a bed wise impedance meter for bed wise void measurement were developed. These probes and meters were calibrated and special testing experiment was conducted.

	Their response characteristics were analyzed from experiment data. The impedance meters are proved to be economical and efficient. Some accomplishments on instrument include the following. (1) Flush mounted film thickness was developed which was mounted on the 6 mm sphere (packed bed particle). This probe can measure film thickness from 50 micrometer to 1 mm. (2) A parallel wire film thickness probe was developed that could measure the film thickness from 1 mm to 3 mm on the spherical particles. (3) A impedance meter was developed which can be mounted external to the packed bed. These meters can measure pipe cross sectional averaged void fraction, detect flow regimes in packed bed, and measure pulse frequencies in two-phase flow in packed bed.
Bibliography Type:	Description: (Last Updated: 12/17/2007)
Dissertations and Theses	Kong X, Revankar ST. "Concurrent Gas-Liquid Two-Phase Down Flow Through A Packed Bed." MS Thesis, School of Nuclear Engineering Purdue University, August 2004. , Aug-2004
Presentation	RevankarS. T.;Kong X. "Study Of Two-Phase Flow Through Packed Bed In Microgravity" Conference-Workshop on Strategic Research to Enable NASA’s Exploration Mission, Glenn Research Center, NASA, Cleveland OH Jun-2004